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ABSTRACT

In today's hypercompetitive customer-centric marketplace, every enterprise strives to gain competitive advantage through customer loyalty, high customer satisfaction and low customer turnover. Through the use of a combination of tools such as analytic technologies, and data-mining techniques and access to real-time data, companies are now able to place a greater emphasis on customer engagement and satisfaction. Today's increased enforcement of standards and stricter legal compliance rules have led call centers to take proactive steps to ensure that enforcement is in compliance with regulations through the use of speech analytics. In the realm of analytic technologies, speech analytics is quickly becoming one of the most demanded technologies in customer engagement optimization and the fastest growing technology in call centers. Organizations have been searching for

ways to translate this wealth of information into holistic, accurate, and actionable insights. The nascent technology is increasingly in more demand as its features become more relevant for call centers, and as organizations seek to capture the voice of the customer (VoC), or the customers' wants and needs, and improve first call resolution (FCR) through post-call and real-time solutions. Speech analytics is a complementary approach for organizations driving optimization, personalization and targeting across their digital channels. Since the latest real-time voice/speech analytics technology can mine 100% of the company's voice contacts, organizations can now be much more successful in capturing and tagging the reasons for the customers' call. Given today's ubiquitous computing trends, such as mobile voice and digital data stream, post-call and real-time applications are solutions to capture the voice of the customer, and improve operations and efficiency, customer experiences, and improve customer loyalty.

Keywords: speech analytics; text mining; data insights; call analytics; customer engagement optimization; use cases; artificial intelligence.

1

Introduction

In today's hypercompetitive business environment, enterprises may strive to gain a sustainable competitive advantage by achieving the highest marks on the most influential elements of customer service: establishing brand and customer loyalty, high customer satisfaction, and a high customer retention rate together with low customer attrition rate. With access to real-time data, companies continually pursue opportunities to achieve this competitive edge and increase the likelihood of success. Tapping into such opportunities requires improving the understanding of customers' perceptions toward the company's brand image, products, and services through a variety of tools such as big data, analytics technologies, and data-mining techniques. For example, call center management has taken steps to implement integrated phone systems designed to improve the quality of its call center agents through insights identified from speech analytics.

In the early days of speech analytics, the adopters were unclear on how to interpret the results from speech analytics technology as there were no actionable insights or results from reports of repeated words or phrases. Around the early 2000s, speech analytics quickly became more widespread as organizations in a variety of industries demanded its

implementation within call centers to maximize customer engagement and customer relationships. Thanks to technological advancements in speech analytics software and an ever-growing data repository, today, companies can uncover business insights and analyze customer perceptions, emotions, and tones as never before. These valuable insights are discovered within a wide range of customer interactions that occur in a variety of customer service channels such as telephone conversations, in-person communication, or customer support channels (e.g., chat, app, text, social media, or email). In Aberdeen Group's benchmark report, "Speech Analytics: Listen to your Customers", companies reported that nearly two-thirds of interactions with customers in call centers involved voice (www.aberdeen.com). The combination of speech analytics technology and big data has garnered the attention of practitioners and academicians alike. Big data components allow call centers and corporations to process and analyze massive amounts of data and translate them into valuable and actionable goals (Kappelman *et al.*, 2014; Moore, 2014).

Over the past decades, numerous research studies have been conducted in the areas of customer engagement and customer relationship management, illustrating the roles and activities performed by customers at various touchpoints. This approach has led firms to strive for a better understanding of the context shaped by the cultural, social, and relational dimensions (Brodie *et al.*, 2011; Chan, 2005). In the realm of analytics technologies, speech analytics is quickly becoming one of the most in-demand technologies in customer engagement optimization and the fastest-growing technology in call centers. Despite being a nascent technology, speech analytics is already increasingly in high demand as its features become more relevant and crucial not only for call centers but also for organizations that seek to capture the voice of the customer (VoC), also known as the customers' wants and needs. The VoC describes the in-depth process of capturing the customer's expectations, preferences, and aversions. Similar to VoC, first call resolution (FCR) is another important facet in achieving customer satisfaction. This is where call center agents successfully listen to customer needs and provide a resolution to solve the customer's issue on the first call. The speech analytics technology is instrumental for successful FCR, through

post-call and real-time solutions by conducting a root-cause analysis, as it investigates the reasons why customers would contact a call center and utilize that information to remedy specific problems at the source.

Given today's ubiquitous computing trends, such as mobile voice and digital data stream, the voice of the customer drives continuous improvement cycles in operational improvement and efficiency, and customer experiences. As identified in the previous researches, voice and speech analytics fall under the umbrella of digital data streams (DDS), which harnesses real-time or near real-time data (Piccoli *et al.*, 2015). Piccoli and Pigni's survey of CIOs demonstrated that there exist exponential untapped opportunities to exploit DDS (Piccoli and Pigni, 2013). While currently used within organizations for monitoring purposes, the combination of a DDS with predictive analytics has the potential to revolutionize customer service and also efficiently improve call center agents' performance.

Speech analytics utilizes speech recognition, predictive analytics, and authentication of the data streams (in this case, the voice of the customer) while assessing customers' complaints in real-time. Assessment occurs through the collection and analysis of current data mixed with historical facts to determine patterns and to predict trends. In the current research, the authors have chosen to focus primarily on speech analytics, serving as an umbrella term encompassing speech analytics, audio-mining technologies. The use of speech analytics typically refers to a broader range of speech products, such as analyzing voice identification, emotion detection, and phonetics/speech analysis.¹ Due to the complexity of vocal elements of a speaker's behavior, the current research excludes in-depth analysis of speech analytics (i.e., syllable emphasis, tone, pitch, tempo, etc.) and focuses primarily on speech analytics. This paper proceeds with the discussion of an overview of enterprise needs for speech analytics, a brief history of the speech recognition, the infrastructure of phonetic versus transcription approaches and real-time versus post-call solutions, major speech analytics vendors and their features, applications found within case studies, and recommendations and guidance. The primary goal of this paper is to help

¹www.rankminer.com/post/voice-analytics-vs-speech-analytics-difference.

business decision-makers educate themselves on the burgeoning field of speech analytics as well as to understand how it impacts the broader enterprise landscape.

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